

DEVENDRA GARWA

(+91) 8058708417 | garwadevendra90@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

IIT Roorkee graduate with expertise in **Machine Learning**, **Predictive Modeling**, and **Data Analytics**. Experienced in developing end-to-end analytical solutions across logistics, credit risk, customer analytics, inventory optimization and geophysical data processing using **Python**, **SQL**, **Power BI**, and modern **ML** techniques.

EDUCATION

Indian Institute of Technology Roorkee (IIT Roorkee) Nov 2021 - May 2026
Integrated M.Tech in Geological Technology

Matrix High School, Sikar 2020
Intermediate (Class XII)

SKILLS

Programming Languages: Python, SQL
Libraries & Packages: Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, CatBoost
Visualization & BI Tools: Power BI, Tableau, Matplotlib, Seaborn, Plotly
Databases & Tools: MySQL, Git, GitHub, Jupyter Notebook, Google Colab, MS Excel
Techniques: Exploratory Data Analysis, Feature Engineering, Classification, Regression, Risk Scoring, SHAP Explainability

EXPERIENCE

Machine Learning Engineer Intern | Technocolabs Softwares Inc. Jun - Jul 2025

- Developed a predictive delivery time estimation model by analyzing **35K+** logistics records using Python, Pandas, and NumPy.
- Engineered **20+** domain-specific features on route efficiency, delay patterns, and shipment metadata to improve predictability.
- Evaluated Lasso, Random Forest, XGBoost, and CatBoost; finalized **CatBoost** achieving an **R² of 0.79** on held-out test set.

PROJECTS

Credit Card Default Prediction | *Classification & Risk Modeling*

- Built a credit risk model on **25K+** records with **30+** features to identify potential loan defaulters using supervised classification.
- Benchmarked Logistic Regression, Random Forest, XGBoost, LightGBM; finalized **XGBoost** with **0.72** F2-score and **75%** recall.
- Employed **SMOTE** for class imbalance and calibrated a **0.27** decision threshold from PR curve for high-risk account flagging.

Flight Delay Analysis and Prediction | *Regression & SHAP Explainability*

- Performed EDA and feature engineering on **30K+** US flight records across **150+** airports to quantify delay patterns.
- Trained **Random Forest** for delay classification (**AUC - 0.94**) and duration regression (**MAE - 6.3 min**) using Scikit-learn.
- Applied **SHAP** to identify carrier and late-aircraft delays as key controllable factors guiding operational improvements.

Oil & Gas Zone Detection Using Seismic Attenuation Analysis | *Geophysical Data Processing & Python*

- Constructed an end-to-end Python pipeline to process **~170 km+** of PSTM 2D seismic data for attenuation analysis.
- Implemented Spectral Ratio and Centroid Frequency Shift methods; cross-validated at tie-line intersections with **<2%** error.
- Isolated anomaly clusters via **Cole-Cole dispersion inversion**, flagging potential hydrocarbon zones on interactive **3D** maps.

Inventory Optimization | *SQL Analytics & Power BI Dashboard*

- Designed a normalized **MySQL** database schema covering **5K+** SKUs across multiple stores to centralize inventory data.
- Defined KPIs using advanced SQL (JOINS, CTEs, window functions) for stock levels, aging inventory, and turnover analysis.
- Created interactive **Power BI** dashboards to visualize SKU, category, and regional inventory trends for real-time monitoring.

Customer Churn Prediction & Retention Strategy | *Classification & SMOTE-ENN*

- Fitted a **Random Forest** classifier on **7,000+** telecom records with a full preprocessing pipeline for encoding and imputation.
- Utilized **SMOTE-ENN** resampling with **GridSearchCV** hyperparameter tuning to improve minority-class recall and robustness.
- Identified top churn drivers via feature importance, informing targeted retention strategies for high-risk customer segments.

LEADERSHIP & EXTRACURRICULARS

Joint Secretary, Public Relations | Rasaionix, IIT Roorkee Jul 2024 - Jun 2025

- Led PR campaigns across campus and social channels, coordinating media partnerships to boost event attendance.
- Managed content calendar and outreach analytics, optimizing messaging performance across communication platforms.

Executive Member | Himalayan Explorers' Club, IIT Roorkee Jun 2023 - May 2024

- Organized logistics for **10+ trekking expeditions**, fostering strong team participation and engagement.
- Delivered safety training to **100+ students**, enhancing readiness and boosting trek participation by **30%**.

CERTIFICATIONS & COURSES

Google Data Analytics (Coursera) | Data Analytics Job Simulation - Forage (Quantium) | WorldQuant BRAIN Challenge - Bronze (2022) | Convolve 4.0 - pan-IIT AI/ML Hackathon (2026) | Winter Consulting - C&A Club, IIT Guwahati | Google Project Management (Coursera)